

Signed homothetic coverings: two planar classes

arXiv:2201.08823, Conjecture 4 (G. Fejes Tóth)

LITERATURE-IMPLIED ANSWER — PARTIAL SUBCASE

Source question. Let $K \subset \mathbb{R}^d$ be a convex body. If translates of $\lambda_i K$, with $-1 < \lambda_i < 1$, cover K , must

$$\sum_i |\lambda_i| \geq d?$$

This is Conjecture 4 on page 6 of Ambrus’s paper.

Exact result identified. The answer is affirmative in each of the following complete planar classes:

1. every nondegenerate triangle;
2. every centrally symmetric planar convex body.

The first class follows from a mixed-orientation theorem of Dumitrescu–Jiang that is not cited in the source discussion. The second follows by reducing signs to the positive planar theorem already cited there. No claim is made for a general noncentrally symmetric planar body or for $d \geq 3$.

Proof intuition. A signed homothet of a triangle is still a triangle with the same three edge directions; a negative coefficient merely reverses its orientation. Dumitrescu–Jiang’s extend/shrink argument was designed for both orientations and converts any such cover into three vertex triangles without increasing the total side length. Their final three-triangle geometry forces total length 2. Affine maps preserve signed homothety coefficients, so the equilateral result is already the result for every triangle.

The decisive literature theorem. Dumitrescu and Jiang, Theorem 1, prove that if a unit equilateral triangle T is covered by finitely many smaller equilateral triangles T_i whose sides are parallel to those of T , and t_i is the side length of T_i , then

$$\sum_i t_i \geq 2.$$

Their definitions immediately below the theorem distinguish positive and negative homothetic copies, and the proof explicitly handles both cases. Thus $t_i = |\lambda_i|$ and this is precisely the signed conjecture for an equilateral triangle.

$n \geq 7$. In particular $U(4) \leq 2$ follows. They also found that $U(3) \leq 2$, and $U(5) < 9/4$. It should be noted here that the inequality $U(k) \leq 2$ for some $k \geq 3$ does not imply for instance $U(k+1) \leq 2$, so in particular having $U(3) \leq 2$ does not imply a similar bound for larger n ; this is so because the values $U(n)$ are defined for each specific value of n .

For the opposite direction, Zhang and Fan conjectured that $U(n) \geq 2$ holds for each $n \geq 3$. Here we prove that $\sum_{i=1}^n t_i \geq 2$ holds for any covering that meets conditions (i) and (ii) above; therefore $U(n) \geq 2$ for each $n \geq 3$. We also improve on all the above-mentioned upper bounds (for $n \geq 5$) and thus obtain exact bounds on $U(n)$ for every $n \geq 3$:

Theorem 1 *For any covering of T with $n \geq 3$ triangles, which satisfies conditions (i) and (ii), we have $\sum_{i=1}^n t_i \geq 2$. This bound is best possible, i.e., for each $n \geq 3$, we have $U(n) = 2$.*

We can similarly define coverings of the unit square with homothetic smaller squares and ask the same question. It turns out that the answer is the same, but the proof is much simpler; we give the details in the last section.

Definitions. A *translate* of a set $X \subset \mathbb{R}^d$ is a set $X + t$, with $t \in \mathbb{R}^d$. A (positive) *homothetic copy* is a scaled translate $\lambda X + t$, for $\lambda > 0$. A *negative homothetic copy* is a scaled translate $\lambda X + t$, for $\lambda < 0$.

2 Proof of Theorem 1

Figure 1: Theorem 1 and the positive/negative definitions in the official EJC PDF.

The two bridge proofs

1. Every planar triangle. Let K be a nondegenerate triangle and let $A(x) = Lx + b$ be an affine bijection taking K to a unit equilateral triangle T . For every scalar λ and translation vector t ,

$$A(\lambda K + t) = \lambda A(K) + (Lt + (1 - \lambda)b).$$

Consequently a cover $K \subset \bigcup_i (\lambda_i K + t_i)$ is carried to a signed homothetic cover of T with exactly the same coefficients. Discard coefficients 0: the union of the remaining finitely many closed homothets contains the dense set obtained by deleting the finitely many zero-homothet points from T , hence contains all of T . Dumitrescu–Jiang now gives

$$\sum_i |\lambda_i| \geq 2 = d.$$

2. Every centrally symmetric planar body. Suppose K is centrally symmetric about c , so $-K = K - 2c$. If $\lambda = -\mu < 0$, then

$$\lambda K + t = -\mu K + t = \mu K + (t - 2\mu c).$$

Thus every negative homothet is a translate of a positive homothet of the same magnitude. Replacing all negative members this way turns the signed cover into a positive homothetic cover with coefficients $|\lambda_i|$. The positive Soltan–Vásárhelyi theorem in dimension two (summarized on the same source page) yields $\sum_i |\lambda_i| \geq 2$.

Why this is only a partial answer. For a noncentrally symmetric planar body with more than three support directions, reflection is not a translate and the triangle’s finite extend/shrink reduction has no known analogue. Exact linear-programming searches during this investigation found no counterexample among triangles, asymmetric trapezoids, or pentagons with two to four mixed-sign homothets and total coefficient 1.8–1.9, but those finite searches do not settle the general case.

$$U_1 + \dots + U_n \subset (1 + \varepsilon) \frac{\text{diam}(K)}{d} T^u,$$

which shows that (8) must hold true. □

Based on Theorem 3, G. Fejes Tóth [FT22+] formulated the following extension of Soltan’s conjecture to coefficients of arbitrary sign.

Conjecture 4 (G. Fejes Tóth). *Assume that $K \in \mathcal{K}^d$ and that $\lambda_1 K, \dots, \lambda_n K$ permit a translative covering of K with $\lambda_i \in (-1, 1)$ for every i . Then*

$$\sum_{i=1}^n |\lambda_i| \geq d.$$

4. ACKNOWLEDGEMENTS

Figure 2: The end of Ambrus’s Theorem 3 proof and Conjecture 4 on source page 6.

References

1. G. Ambrus, *A generalization of Bang’s lemma*, Proc. Amer. Math. Soc. 151 (2023), 1277–1284; arXiv:2201.08823. Conjecture 4, p. 6.
2. A. Dumitrescu and M. Jiang, *On a covering problem for equilateral triangles*, Electron. J. Combin. 15 (2008), R37. Theorem 1 and definitions, pp. 1–2.
3. V. P. Soltan and È. Vásárhelyi, *Covering a convex body by smaller homothetic copies*, Geom. Dedicata 45 (1993), 101–113.